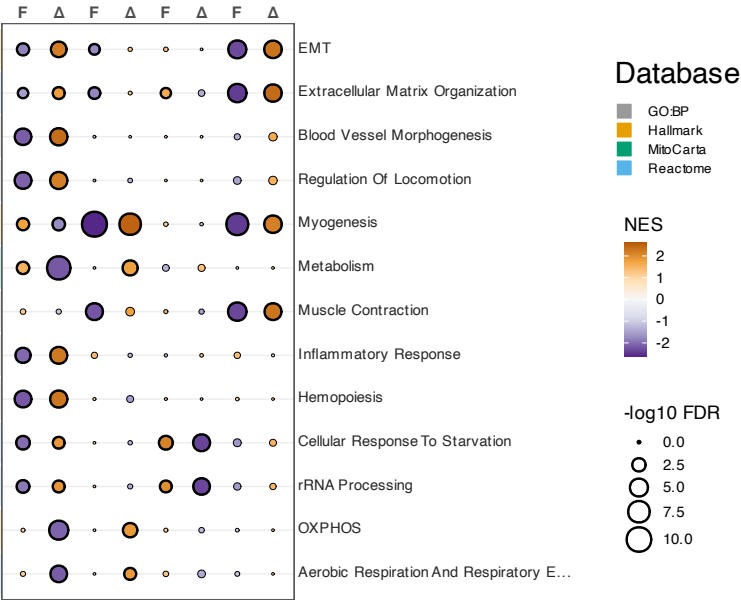


a

Pathways whose link to phenotype differs by sex

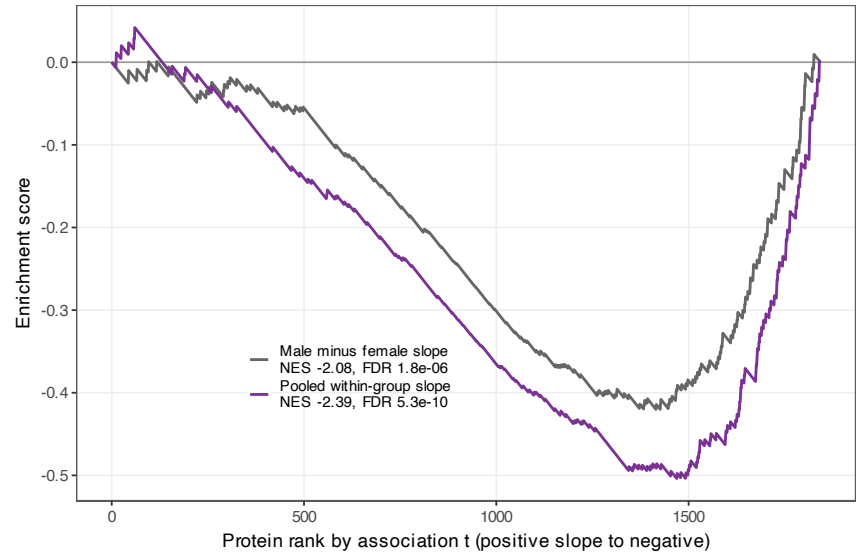
Proteins ranked by phenotype t, fgsea; columns in pairs, F (female fit) then Δ (male minus female slope): soleus pyruvate, capillary density, CSA, soleus octanoyl; black ring: FDR ≤ 0.05



c

Oxidative phosphorylation against soleus pyruvate conductance

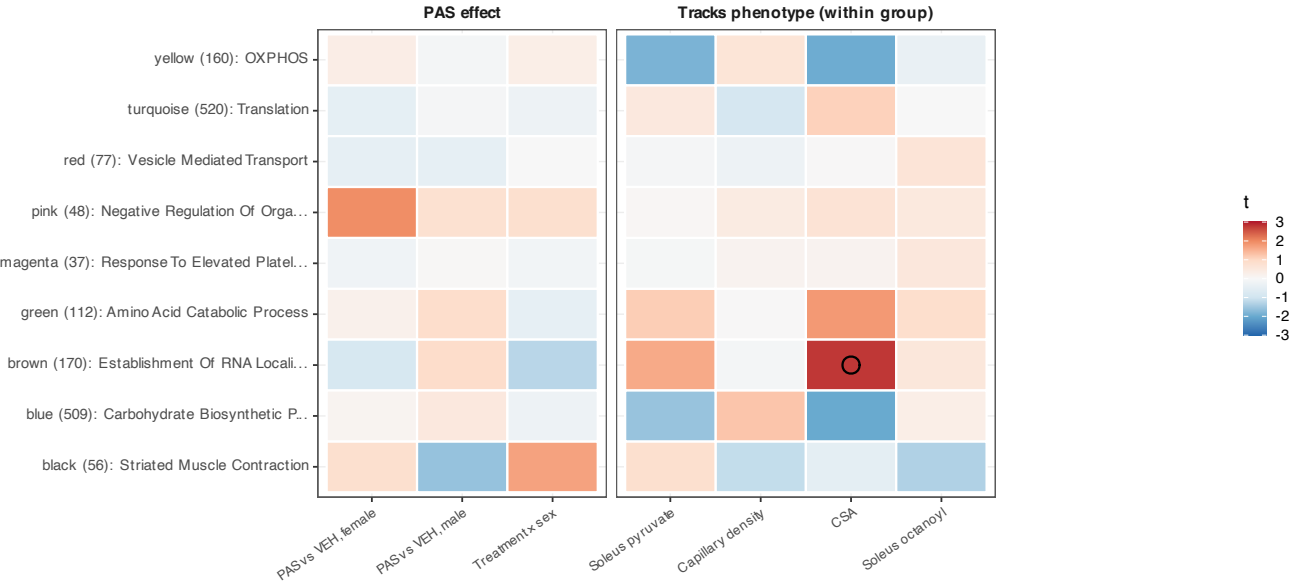
fgsea running score; Hallmark OXPHOS in the protein ranking



b

WGCNA modules: identity, PAS effect, phenotype link

Module (proteins): top annotated term; t from emmeans contrasts or within-group slope; ring: p ≤ 0.05



d

Proteins behind the sex-different pathways

Leading edge (first 5) of the 6 strongest interaction terms; PAS log2 FC per sex; * p ≤ 0.05

